

A Brief History of Vaccination - World Health Organization

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Learn the story of these life-saving jabs

For centuries, humans have looked for ways to protect each other against deadly diseases. From experiments and taking chances to a global vaccine roll-out in the midst of an unprecedented pandemic, immunization has a long history.

Vaccine research can raise challenging ethical questions, and some of the experiments carried out for the development of vaccines in the past would not be ethically acceptable today. Vaccines have saved more human lives than any other medical invention in history.

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Scroll on to take a journey through the last millennium to see how these extraordinary discoveries and achievements have changed our lives.

1400s to 1700s

From at least the 15th century, people in different parts of the world have attempted to prevent illness by intentionally exposing healthy people to smallpox— a practice known as variolation (after a name for smallpox, 'la variole'). Some sources suggest these practices were taking place as early as 200 BCE.

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prevent illness by intentionally exposing healthy people to smallpox.

In 1721, Lady Mary Wortley Montagu brought smallpox inoculation to Europe, by asking that her two daughters be inoculated against smallpox as she had observed practice in Turkey.

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In 1774, Benjamin Jesty makes a breakthrough. Testing his hypothesis that infection with cowpox - a bovine virus which can spread to humans - could protect a person from

smallpox

1796

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Dr Edward Jenner created the world's first successful vaccine. He found out that people infected with cowpox were immune to smallpox.

In May 1796, English physician Edward Jenner expands on this discovery and inoculates 8-year-old James Phipps with matter collected from a cowpox sore on the hand of a milkmaid. Despite suffering a local reaction and feeling unwell for several days, Phipps made a full recovery.

Two months later, in July 1796, Jenner inoculates Phipps with matter from a human smallpox sore in order to test Phipps' resistance. Phipps remains in perfect health, and becomes the first human to be vaccinated against smallpox. The term 'vaccine' is later coined, taken from the Latin word for cow, vacca.

Read more about the history of Smallpox vaccination.

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1806

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French Emperor Napoleon Bonaparte and American President Thomas Jefferson acknowledge Dr Edward Jenner's work and endorse the smallpox vaccine.

The 1800s

In 1872, despite enduring a stroke and the death of 2 of his daughters to typhoid, Louis Pasteur creates the first laboratory-produced vaccine: the vaccine for fowl cholera in chickens.

In 1885, Louis Pasteur successfully prevents rabies through post-exposure vaccination. The treatment is controversial. Pasteur has unsuccessfully attempted to use the vaccine on humans twice before, and injecting a human with a disease agent is still a new and uncertain method.

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Louis Pasteur successfully prevents rabies through post-exposure vaccination.

Pasteur is not a medical doctor. But, despite the risk, he begins a course of 13 injections with patient Joseph Meister, each containing a stronger dose of the rabies virus. Meister survives and later becomes the caretaker of Pasteur's tomb in Paris.

In 1894, Dr Anna Wessels Williams isolates a strain of the diphtheria bacteria that is crucial in the development of an antitoxin for the disease.

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The “Spanish Flu” pandemic kills 1 in 67 United States soldiers, making an influenza vaccine a US military priority.

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The 1900s

From 1918 to 1919, the Spanish Flu pandemic kills an estimated 20-50 million people worldwide, including 1 in 67 United States soldiers, making an influenza vaccine a US military priority. Early experiments with influenza vaccines are carried out: the US Army Medical School tests 2 million doses in 1918, but results are inconclusive.

Read more about the history of Influenza vaccination.

In 1937 Max Theiler, Hugh Smith and Eugen Haagen develop the 17D vaccine against yellow fever. The vaccine is approved in 1938 and over a million people have received it that year. Theiler goes on to be awarded the Nobel Prize.

In 1939, bacteriologists Pearl Kendrick and Grace Eldering demonstrate the efficacy of the pertussis (whooping cough) vaccine. The scientists show that vaccination reduces the rates at which children get sick from 15.1 per 100 children to 2.3 per 100.

By 1945, the first influenza vaccine is approved for military use, followed in 1946 by an approval for civilian use. The research is led by doctors Thomas Francis Jr and Jonas Salk, who both go on to be closely associated with the polio vaccine.

From 1952-1955, the first effective polio vaccine is developed by Jonas Salk and trials begin. Salk tests the vaccine on himself and his family the following year, and mass trials involving over 1.3 million children take place in 1954.

By 1960, a second type of polio vaccine, developed by Albert Sabin, is approved for use. Sabin's vaccine was live-attenuated (using the virus in weakened form) and could be given Orally, as drops or on a sugar cube. The oral polio vaccine (OPV) was first tested and produced in the Soviet Union and Eastern Europe. Czechoslovakia becomes the first country in the world to eliminate polio.

Read more about the history of Polio vaccination.

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1967

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Mass vaccination begins with

the World Health Organization
announcing the Intensified
Smallpox Eradication Programme.

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In 1967, the World Health Organization announces the Intensified Smallpox Eradication Programme, which aims to eradicate smallpox in more than 30 countries through surveillance and vaccination. Eradication means more than the elimination of a disease in a single area - WHO defines it as the "permanent reduction to zero of a specific pathogen, as a result of deliberate efforts, with no more risk of reintroduction".

Smallpox has been mostly eliminated in Western Europe, North America and Japan by this time. Following the announcement, there is unprecedented global solidarity. Despite the ongoing Cold War, the United States and the Soviet Union are united in support of the programme.

In 1969, four years after Dr Baruch Blumberg discovers the hepatitis B virus, he works with microbiologist Irving Millman to develop the first hepatitis B vaccine, using a heat-treated form of the virus.

A plasma-derived inactivated vaccine is approved for commercial use from 1981 to 1990, and a genetically engineered (or DNA recombinant) vaccine, developed in 1986, is still in use today.

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1971

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The measles vaccine (1963) is combined with recently developed vaccines against mumps (1967) and rubella (1969) into a single vaccination (MMR).

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In 1971 the measles vaccine (1963) is combined with recently developed vaccines against mumps (1967) and rubella (1969) into a single vaccination (MMR) by Dr Maurice Hilleman.

Read more about the history of measles vaccination.

In 1974 the Expanded Programme on Immunization (EPI, now the Essential Programme on Immunization) is established by WHO to develop immunization programmes throughout the world. The first diseases targeted by the EPI are diphtheria, measles, polio, tetanus, tuberculosis and whooping cough.

In 1978 a polysaccharide vaccine that protects against 14 different strains of pneumococcal pneumonia is licensed, and in 1983 it is expanded to protect against 23 strains.

In 1980 the World Health Assembly, acting on recommendation from the WHO Global Commission for the Certification of Smallpox Eradication, declares smallpox eradicated:

“The world and all its people have won freedom from smallpox, which was the most devastating disease sweeping in epidemic form through many countries since earliest times, leaving death, blindness and disfigurement in its wake.”

From 1970s to 1980s in the USA, whooping cough cases hit an all-time low in 1976. But the success of the pertussis vaccine is hampered by a decline in uptake: with so few whooping cough cases, fears about rare but serious side effects of the whole-cell vaccine start to outweigh fears of the disease itself.

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licensed, after David H Smith founds a company to produce it. Smith and Porter W Anderson Jr had been working together on a vaccination since 1968

1988

WHO launches the Global Polio Eradication Initiative.

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In 1988 following the eradication of smallpox, WHO sets its sights on poliomyelitis, launching a Global Polio Eradication Initiative. In the late 1980s, polio is endemic in 125 countries, and the initiative aims to achieve its eradication by the year 2000.

By 1994, polio is eradicated from the Americas, followed by Europe in 2002, and by 2003 the disease is endemic in just 6 countries. The effort continues.

In 1995 Anne Szarewski leads a team who outline the role of human papillomavirus (HPV) in cervical cancer detection and screening, and researchers begin work on an HPV vaccine.

HPV viruses are very common, often with minimal symptoms, but high-risk HPV strains can go on to cause other medical conditions, particularly cervical cancer. Szarewski goes on to be principal investigator in the development of the bivalent HPV vaccine.

In 1999 the first vaccine against rotavirus, the most common cause of severe diarrhoeal disease in young children, is withdrawn only a year after it was approved, due to concerns about the risk of intestinal problems. A lower-risk version of the vaccine is introduced in 2006. It takes until 2019 for it to be in use in over 100 countries.

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In 2006 the first vaccine for Human Papillomavirus (HPV) is approved. HPV vaccination goes on to become a key part of the effort to eliminate cervical cancer.

The success of the Meningitis Vaccine Project highlights the key role public-private partnerships can play in helping to develop vaccines.

In 2016 the success of the Meningitis Vaccine Project highlights the key role public-private partnerships can play in helping to develop vaccines. In its first 5 years of use, the vaccine has nearly eliminated serogroup A meningococcal disease in meningitis belt countries of Africa, and it is now being integrated into routine national immunization programmes.

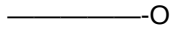
The World Health Assembly welcomes the R&D Blueprint, a global strategy and preparedness plan that allows the rapid activation of research and development activities during epidemics. Its aim is to fast-track the availability of effective tests, vaccines and medicines that can be used to save lives and avert large-scale crises.

Following years of accelerated vaccinations, the Americas region is declared free of endemic measles. Outbreaks in several countries, caused by gaps in vaccination coverage, see the disease begin to reemerge in 2018. WHO and PAHO increase surveillance and launch vaccination campaigns.

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2019



WHO prequalifies an Ebola vaccine for use in countries at high risk.

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In 2019, the malaria vaccine pilot implementation is launched in Ghana, Malawi and Kenya. The RTS/S vaccine is the first vaccine that can significantly reduce the deadliest and most prevalent strain of malaria in young children, the group at highest risk of dying from the disease.

WHO prequalifies an Ebola vaccine for use in countries at high risk, as part of a broader set of tools in response to the disease. In 2021 a global vaccine stockpile is established to ensure outbreak response.

A third-generation smallpox vaccine is approved for prevention of monkeypox, thus becoming the first monkeypox vaccine.

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2021



WHO calls on Member States to prioritize vaccination against COVID-19 of health workers and at-risk groups in all countries.

On 30 January, 2020 the WHO Director General declares the outbreak of novel coronavirus 2019 (SARS-CoV-2) to be a Public Health Emergency of International Concern. On 11 March, WHO confirms that COVID-19 is a pandemic.

Effective COVID-19 vaccines are developed, produced and distributed with unprecedented speed, some using new mRNA technology. In December 2020, just 1 year after the first case of COVID-19 was detected, the first COVID-19 vaccine doses are administered.

In 2021 the COVID-19 vaccine roll-out continues, with doses delivered and administered across continents. But efforts to curb the pandemic are threatened by inequities in vaccination coverage: as of July 2021, almost 85% of vaccines have been administered in high- and upper-middle-income countries, and over 75% have been administered in only 10 countries alone.

WHO calls on Member States to prioritize vaccination of health workers and at-risk groups in lower-income countries, in order to stop severe disease and death, keep health workers

safe and reopen societies and economies.

For over 2 centuries, people have been vaccinated against deadly diseases, ever since the world's first vaccine was devised against smallpox. History has taught us that a full and effective global response to vaccine-preventable diseases takes time, financial support and collaboration - and requires continued vigilance.

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From groundbreaking practices in the 1500s to the new technologies used in COVID-19 vaccines, we have come a long way. Vaccines now help protect against more than 20 diseases, from pneumonia to cervical cancer and Ebola; and in just the last 30 years, child deaths have declined by over 50%, thanks in large part to vaccines. But more must be done.

In many parts of the world, 1 in 5 children still goes unvaccinated. The coming decades will need global cooperation, funding, commitment and vision to ensure that no child or adult suffers or dies from a vaccine-preventable disease.

Explore more at g.co/historyofvaccination.

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